
Chapter 2- Software Process Models

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Reference Books:

- 1) Fundamentals of Software Engineering, Rajib Mall , PHI, Latest edition
- 2) Software Engineering, I. Sommerville, Pearson Education, Asia.

Topics covered

- Software process models
- Software Development Life Cycle (SDLC)
 - SDLC models
 - Waterfall model

Focus of Modern Software Development

- Error prevention than error correction
- Detecting as many errors as possible in each phase of development
- Periodic reviews are carried out after each phase
- Code is only part of the software development
- Code is not the only deliverable

Focus of Modern Software Development

- Systematic
 - Standard design techniques
 - Standard testing techniques
 - Extensive usage of Case tools
- Better visibility of design and code with good quality, consistent and standard documents
- Better fault diagnosis and smoother maintenance

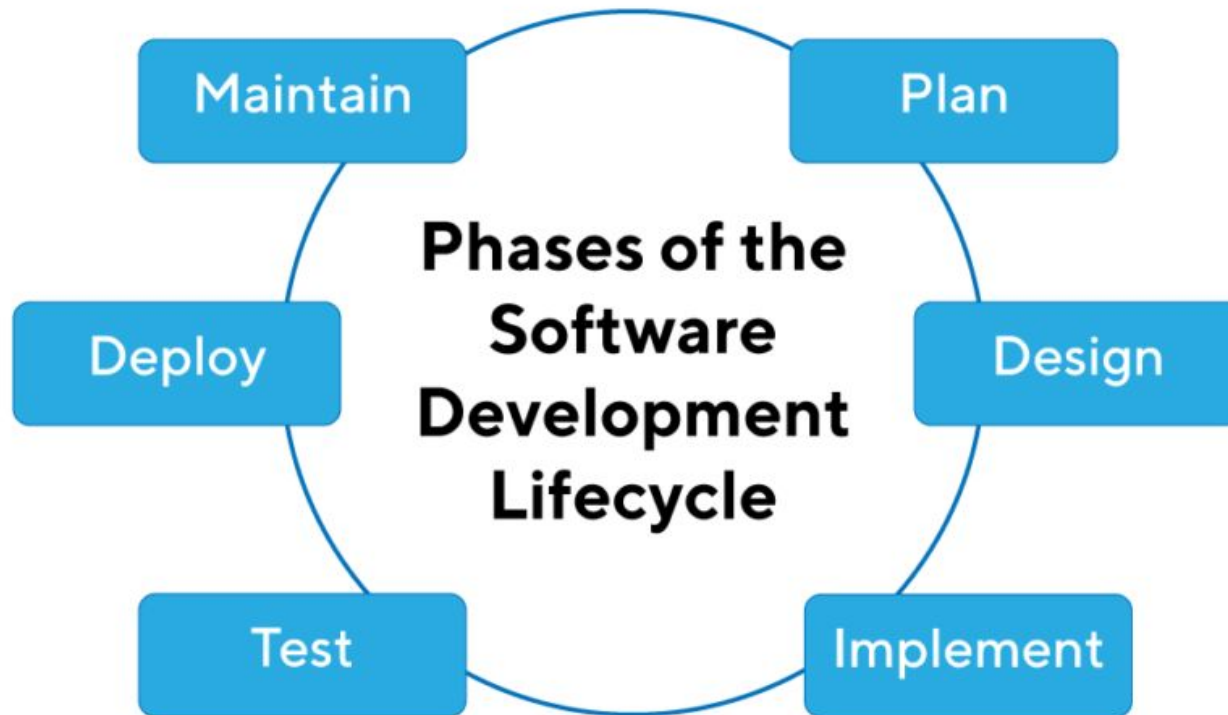
Software Process Models

A software process model is an abstract representation of a process.

Software Process

- **Software specification**, where customers and engineers define the software that is to be produced and the constraints on its operation.
- **Software development**, where the software is designed and programmed.
- **Software validation**, where the software is checked to ensure that it is what the customer requires.
- **Software evolution**, where the software is modified to reflect changing customer and market requirements.

Software Development Life Cycle



Software Development Life Cycle (SDLC) Model

- Describes the different activities that need to be carried out for the software to evolve in its life cycle
- A descriptive and diagrammatic model of software life cycle
- Identifies the activities and establish a precedence ordering among them
- Works phase by phase/stage

Software Development Life Cycle (SDLC) Model

- Every phase can have multiple activities
- Example: The activities of design phase can be – structured design, design review etc.
- Phase descriptions may also include:
 - Products, which are the outcomes of a activity;
 - Roles, which reflect the responsibilities of the people involved in the phase;
 - Pre- and post-conditions, which are statements that are true before and after an activity has been enacted or a product produced.

Software Development Life Cycle (SDLC) Model

Advantages of both graphical and written description

- Helps common understanding of activities among the developers
- Identification of inconsistencies, redundancies, and omissions in the development process
- Helps in tailoring according to specific projects
- Helps in defining milestones
- Avoids 99% complete syndrome

Examples of SDLC model

- Waterfall model
- Evolutionary model
- V model
- Prototyping model
- Spiral model
- **Agile model**

Plan-driven and Agile Processes

- Plan-driven processes are processes where all of the process activities are planned in advance and progress is measured against this plan.
- In agile processes, planning is incremental and it is easier to change the process to reflect changing customer requirements.
- In practice, most practical processes include elements of both plan-driven and agile approaches.
- There are no right or wrong software processes.

Classic Waterfall Model

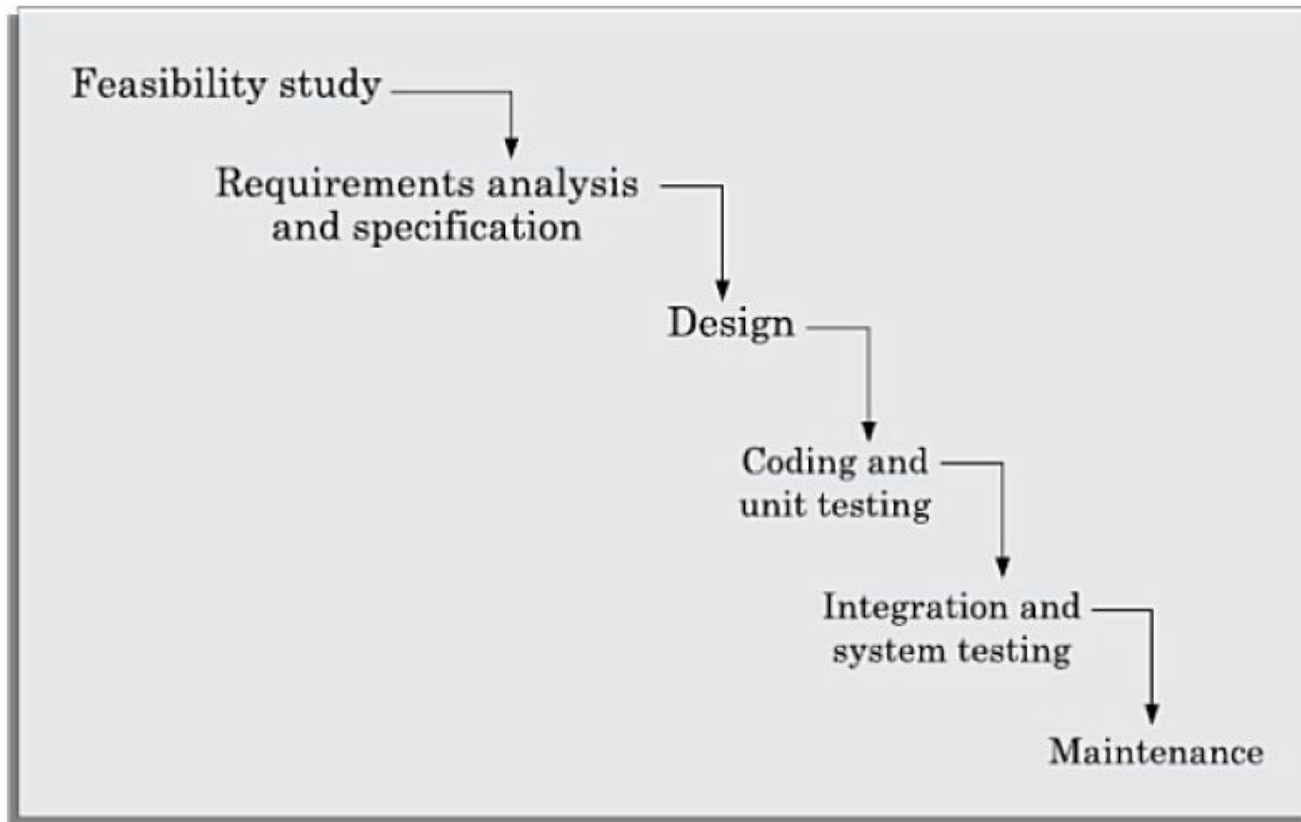


Figure 2.1. A classic waterfall model with its phases

Relative Effort during Phases

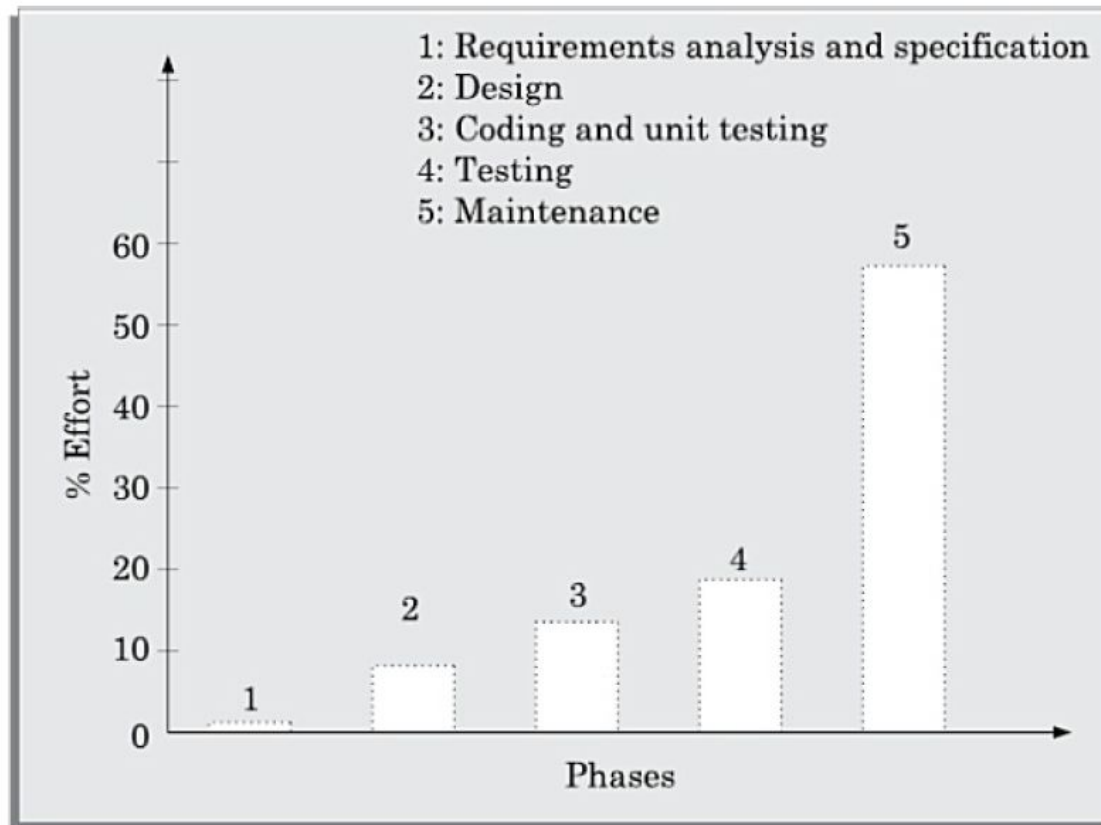


Figure 2.2. Relative effort distribution among different phases of a typical product.

Feasibility Study

1. Economic Feasibility
2. Technical Feasibility
3. Schedule Feasibility

Activities during Feasibility Study

- Workout an overall understanding of the system
- Formulate different strategies
- Examine the feasibility of the different strategies from the three aspects
- Cost (development, operational) benefit (quantifiable, non-quantifiable) analysis
- Writing an affective 'business case'

An effective “Business case”

1. Executive summary
2. Project background
3. Business opportunity
4. Costs
5. Benefits
6. Risks

Drawbacks of Waterfall Model

- Difficulty in accommodating change after the process is underway. In principle, a phase has to be complete before moving onto the next phase.
- Inflexible partitioning of the project into distinct stages makes it difficult to respond to changing customer requirements.

Phase2: Requirement Analysis and Specification

Aim of this phase is to

1. Understand the exact requirements of the customer
2. Document it properly

Activities during Requirement Analysis and Specification

- Requirement gathering
- Analyse the gathered requirements
 - Remove contradictions, ambiguities
- Organizing the requirements
- Document the finalized requirements
 - Software Requirement Specification (SRS) document

Classic Waterfall Model

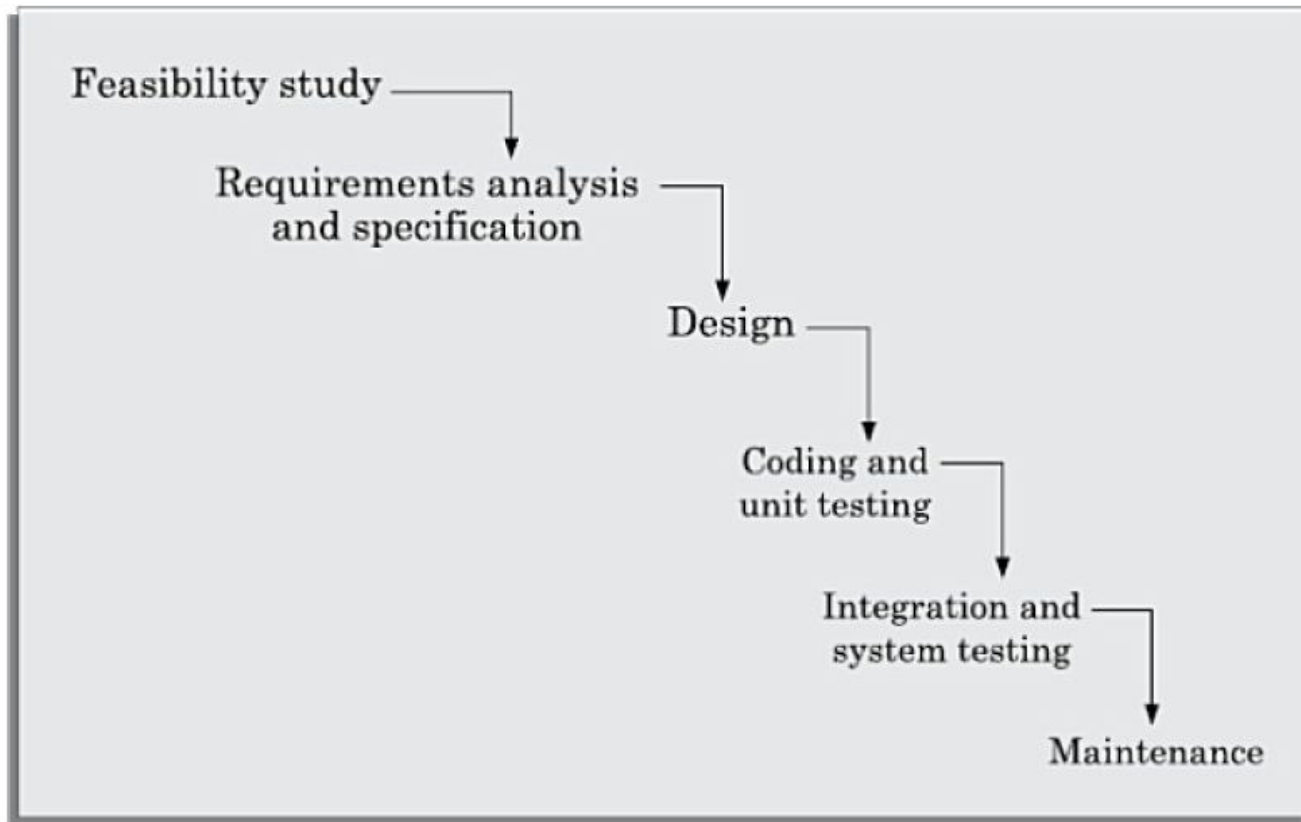


Figure 2.1. A classic waterfall model with its phases

Phase3: Design

- Traditional approach
 1. Structured analysis
 2. Structured design
- Object-oriented approach
 1. Object-oriented analysis
 2. Object-oriented design

Phase3: Design - Structured Analysis and Structured Design (SA/SD)

- Goal is to perform functional decomposition of the system, reduce the risk of system failure
- Transform the specifics of SRS to be easily implemented using conventional programming languages.
- Involves of two phases:
 - Analysis phase: Data Flow Diagram (DFD), Data dictionary, state transition diagram, and ER diagram
 - Design phase: Structure chart, pseudo-code

Phase3: Design – SA/SD

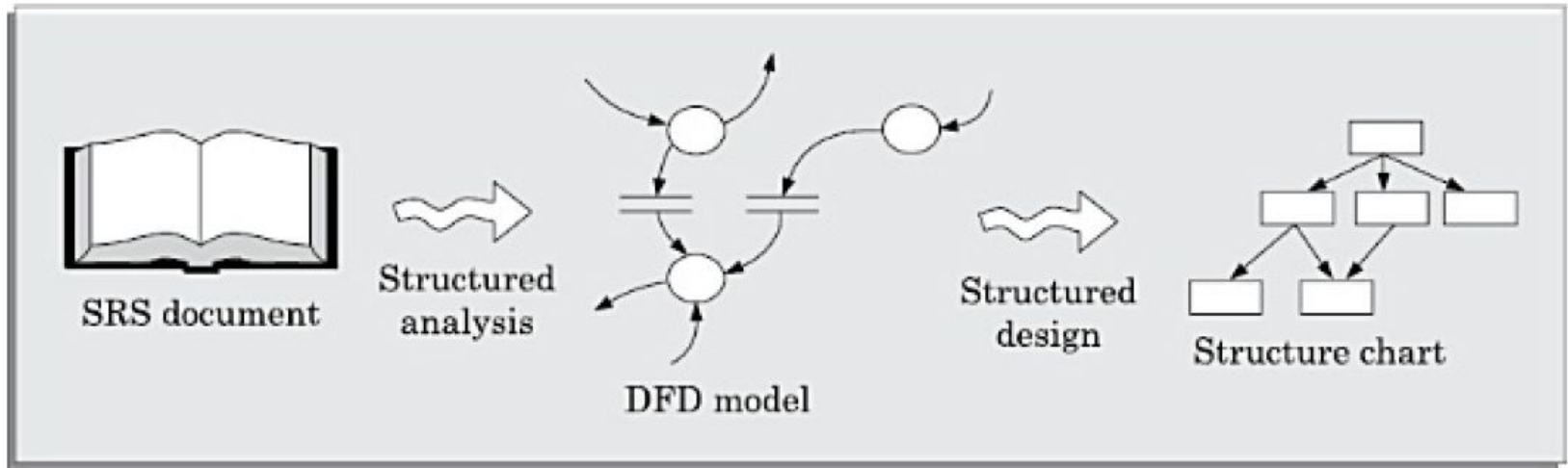


Figure 2.3. Structured analysis and structured design methodology

Phase3: Design – SA/SD -> DFD

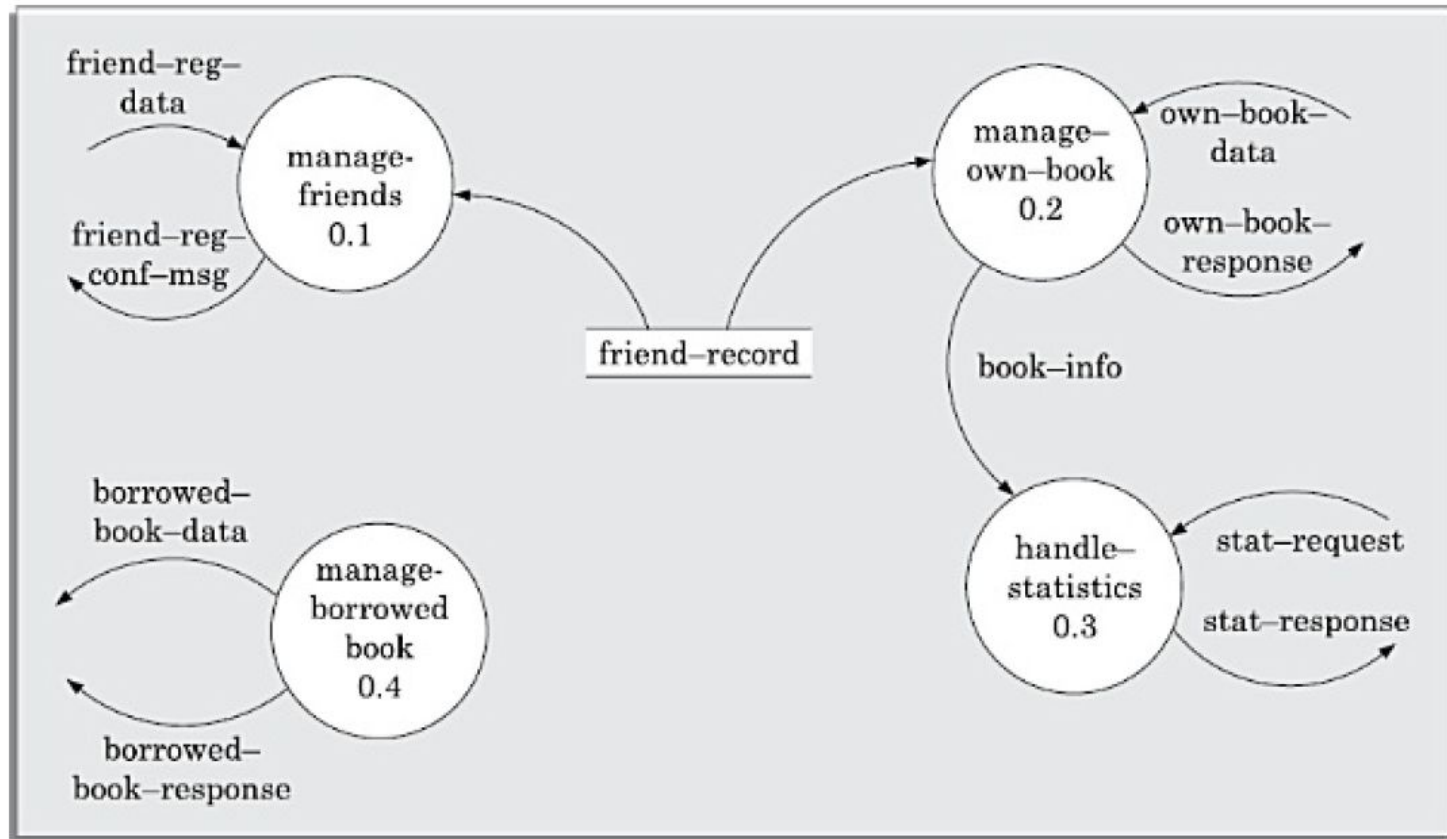


Figure 2.4. Data flow diagram of a personal library management system.

Phase3: Design – SA/SD -> Structure chart

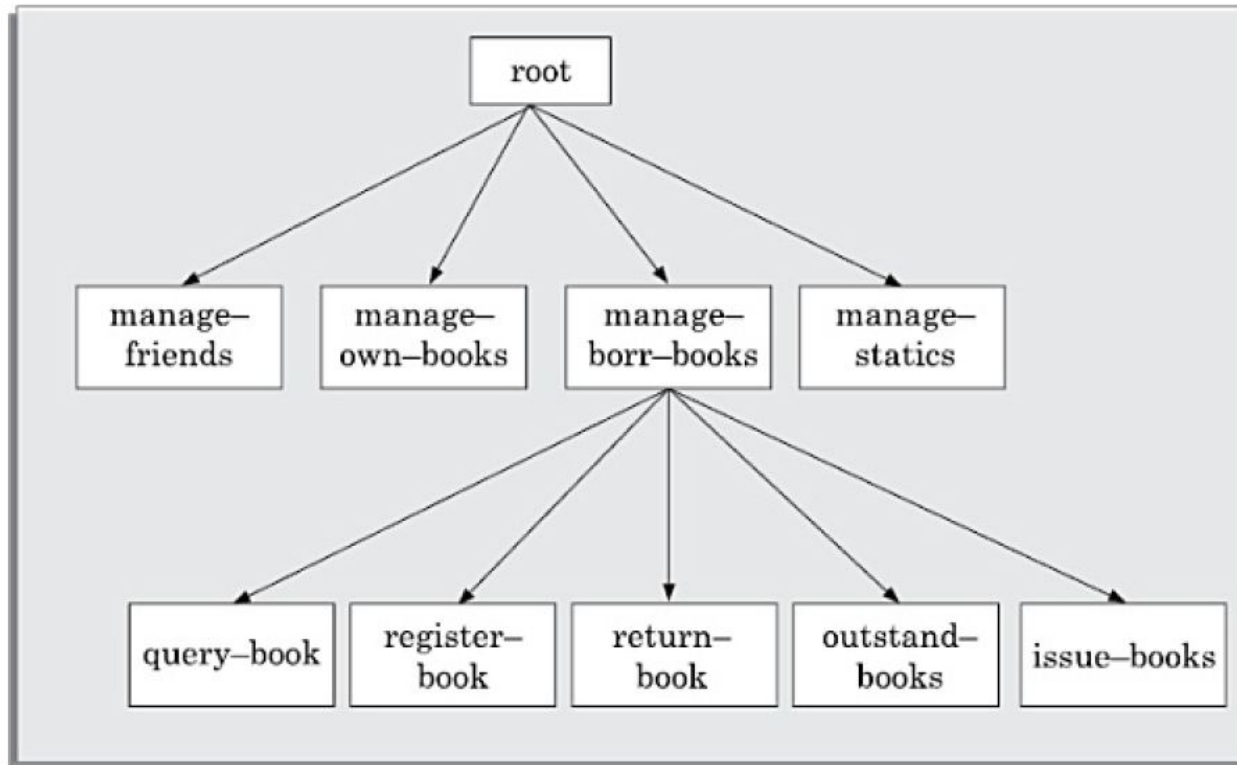


Figure 2.5. Structure chart of a personal library management system.

Phase3: Design – Object-oriented Analysis and Design

- Object-oriented analysis (OOA)
 - Data, functional, behavioural models to model the information domain, represent the behaviour and describe functions.
 - Object modelling techniques (OMT)
- Object-oriented design (OOD)
 - Specifies several levels of modularity
 - Object-Oriented design pyramid

Phase3: Design – Structured vs Object-oriented design

Structured Analysis and Structured Design (SA/SD)	Object oriented Analysis and Design
Focuses on process and procedures of the system	Focuses on real-world objects and data structures
Involves DFD, state-transition diagrams etc.	UML diagrams are used to model
Risk is high, and reusability is low	Risk is less, reusability is high
Extensively used for SDLC models	Generally used for incremental models rather than SDLC models

Classic Waterfall Model

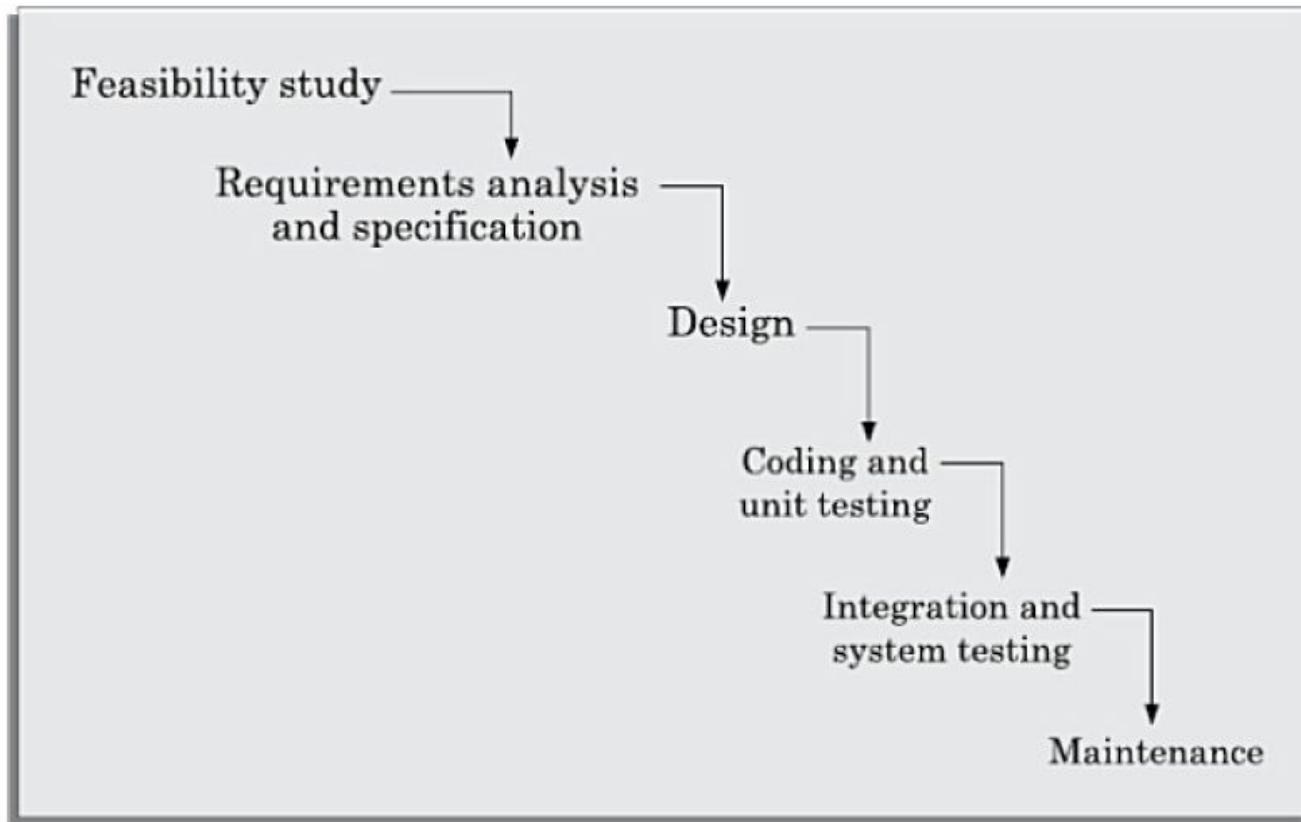


Figure 2.1. A classic waterfall model with its phases

Phase 5: Integration and System Testing

- Ensures that the developed system functions according to the requirements specified in the SRS document.
- Tests are done modules during each level of integration.
- After all successful integration, the final test is done on the complete system.

Phase 6: Maintenance

- Corrective maintenance
 - Fixing errors that were not discovered during the development phases
- Perfective maintenance
 - To add some new functionality
 - Improve the performance
- Adaptive maintenance
 - If there is any change in the environment, e.g. porting to a different OS

Drawbacks of Waterfall Model

- Not realistic
- Difficulty in accommodating change after the process is underway. In principle, a phase has to be complete before moving onto the next phase.
- Inflexible partitioning of the project into distinct stages makes it difficult to respond to changing customer requirements.

Next Session

- Other Software Process Models
 - Derivatives of Waterfall model